KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3092 | 3 Credits: 3 | Program Core | Source-grounded

Polymer Science

 To explain the concept of Tg and crystallinity based on states of phases and

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3092 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3092.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Polymer Technology
Course code	3092
Course title	Polymer Science
Semester	3 Credits: 3
Course category	Program Core
Periods per week	3 (L:2,T:1,P:0) Periods per semester: 45
Periods per semester	45

Item	Official information
Credits	3
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To explain the concept of Tg and crystallinity based on states of phases and
- To illustrate physical and chemical methods of polymer identification
- To provide concept of molecular weight and stereo regularity of polymers.
- To summarize polymer reactions and polymer modifications and its impact on

Course outcomes

CO	Official outcome	Study level
CO1	10 Applying chemical methods of polymer identification Describe the concept of molecular weight and	See official syllabus
CO2	12 Applying experimental method of its determination	See official syllabus
CO3	Illustrate the stereo regularity of polymers 11 Understanding Explain polymer reactions and polymer	See official syllabus
CO4	10 Understanding modifications	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	identification	4	As specified
Module 2	determination	4	As specified
Module 3	Illustrate the stereo regularity of polymers	4	As specified
Module 4	Explain Polymer reactions and polymer modifications	4	As specified

MODULE 1

identification

This module is presented from the official syllabus scope for Polymer Science. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: identification
- Official duration: As specified
- Official topic entries captured: 4

- The full source excerpt is preserved below for coverage checking.

basis of States of phase and aggregates of 3 Understanding polymers. Explain Glass transition temperature (Tg)-

basis of States of phase and aggregates of 3 Understanding polymers. Explain Glass transition temperature (Tg)- is an official topic in Module 1 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify basis of States of phase and aggregates of 3 Understanding polymers. Explain Glass transition temperature (Tg)- using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, basis of states of phase and aggregates of 3 understanding polymers. explain glass transition temperature (tg)- should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What basis of States of phase and aggregates of 3 Understanding polymers. Explain Glass transition temperature (Tg)- means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] basis of States of phase and aggregates of 3 Understanding polymers. Explain Glass transition temperature (Tg)- [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Melting temperature (Tm) -Flow temperature 2 Understanding (Tf) Explain Crystalline polymer and amorphous

Melting temperature (Tm) -Flow temperature 2 Understanding (Tf) Explain Crystalline polymer and amorphous is an official topic in Module 1 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Melting temperature (Tm) -Flow temperature 2 Understanding (Tf) Explain Crystalline polymer and amorphous using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, melting temperature (tm) -flow temperature 2 understanding (tf) explain crystalline polymer and amorphous should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Melting temperature (Tm) -Flow temperature 2 Understanding (Tf) Explain Crystalline polymer and amorphous means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Melting temperature (Tm) -Flow temperature 2 Understanding (Tf) Explain Crystalline polymer and amorphous □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Understanding polymer Explain Polymer identification by chemical and

2 Understanding polymer Explain Polymer identification by chemical and is an official topic in Module 1 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding polymer Explain Polymer identification by chemical and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding polymer explain polymer identification by chemical and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding polymer Explain Polymer identification by chemical and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-[illegible] 2 Understanding polymer Explain Polymer identification by chemical and [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Applying physical methods State of phase and aggregates of polymers and polymer identification Glass transition temperature (Tg) - Melting temperature (Tm) - Flow temperature (Tf). Crystalline polymer - Amorphous polymer - Crystallinity Crystallisability -factors affecting Tg - Dilatometry determination of Tg-X-Ray diffraction method for determination of crystallinity - Comparison of Elastomers, plastics and fibers in terms of Tg and crystallinity. Polymer identification - chemical and physical methods. TGA, DSC and IR spectroscopy. Explain the concept of molecular weight and experimental method of its

3 Applying physical methods State of phase and aggregates of polymers and polymer identification Glass transition temperature (Tg) - Melting temperature (Tm) - Flow temperature (Tf). Crystalline polymer - Amorphous polymer - Crystallinity Crystallisability -factors affecting Tg - Dilatometry determination of Tg-X-Ray diffraction method for determination of crystallinity - Comparison of Elastomers, plastics and fibers in terms of Tg and crystallinity. Polymer identification - chemical and physical methods. TGA, DSC and IR spectroscopy. Explain the concept of molecular weight and experimental method of its is an official topic in Module 1 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying physical methods State of phase and aggregates of polymers and polymer identification Glass transition temperature (Tg) - Melting temperature (Tm) - Flow temperature (Tf). Crystalline polymer - Amorphous polymer - Crystallinity Crystallisability -factors affecting Tg - Dilatometry determination of Tg-X-Ray diffraction method for determination of crystallinity - Comparison of Elastomers, plastics and fibers in terms of Tg and crystallinity. Polymer identification - chemical and physical methods. TGA, DSC and IR spectroscopy. Explain the concept of molecular weight and experimental method of its using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying physical methods state of phase and aggregates of polymers and polymer identification glass transition temperature (tg) - melting temperature (tm) - flow temperature (tf). crystalline polymer - amorphous polymer - crystallinity crystallisability -factors affecting tg - dilatometry determination of tg-x-ray diffraction method for determination of crystallinity - comparison of elastomers, plastics and fibers in terms of tg and crystallinity. polymer identification - chemical and physical methods. tga, dsc and ir spectroscopy. explain the concept of molecular weight and experimental method of its should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying physical methods State of phase and aggregates of polymers and polymer identification Glass transition temperature (Tg) - Melting temperature (Tm) - Flow temperature (Tf). Crystalline polymer - Amorphous polymer - Crystallinity Crystallisability -factors affecting Tg - Dilatometry determination of Tg-X-Ray diffraction method for determination of crystallinity - Comparison of Elastomers, plastics and fibers in terms of Tg and crystallinity. Polymer identification - chemical and physical methods. TGA, DSC and IR spectroscopy. Explain the concept of molecular weight and experimental method of its means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 3 Applying physical methods State of phase and aggregates of polymers and polymer identification Glass transition temperature (Tg) - Melting temperature (Tm) - Flow temperature (Tf). Crystalline polymer - Amorphous polymer - Crystallinity Crystallisability -factors affecting Tg - Dilatometry determination of Tg-X-Ray diffraction method for determination of crystallinity - Comparison of Elastomers, plastics and fibers in terms of Tg and crystallinity. Polymer identification - chemical and physical methods. TGA, DSC and IR spectroscopy. Explain the concept of molecular weight and experimental method of its □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□

□□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
identification
            Generalize concept Tg and crystallinity on the
  M1.01     basis of States of phase and aggregates of           3
Understanding
            polymers.
            Explain Glass transition temperature (Tg)-
  M1.02     Melting temperature (Tm) -Flow temperature           2
Understanding
            (Tf)
            Explain Crystalline polymer and amorphous
  M1.03                                                          2
Understanding
            polymer
            Explain Polymer identification by chemical and
  M1.04                                                          3
Applying
            physical methods

Contents:
State of phase and aggregates of polymers and polymer identification Glass
transition
temperature (Tg) - Melting temperature (Tm) - Flow temperature (Tf). Crystalline
polymer -
Amorphous polymer - Crystallinity Crystallisability -factors affecting Tg -
Dilatometry
determination of Tg-X-Ray diffraction method for determination of crystallinity -
Comparison of Elastomers, plastics and fibers in terms of Tg and crystallinity.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

determination

This module is presented from the official syllabus scope for Polymer Science. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: determination
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

2 Applying micro and macro molecules. Explain different molecular weight of polymers,

2 Applying micro and macro molecules. Explain different molecular weight of polymers, is an official topic in Module 2 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying micro and macro molecules. Explain different molecular weight of polymers, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying micro and macro molecules. explain different molecular weight of polymers, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying micro and macro molecules. Explain different molecular weight of polymers, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 2 Applying micro and macro molecules. Explain different molecular weight of polymers, □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

molecular weight distribution and 3 Applying polydispersity.

molecular weight distribution and 3 Applying polydispersity. is an official topic in Module 2 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify molecular weight distribution and 3 Applying polydispersity. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, molecular weight distribution and 3 applying polydispersity. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What molecular weight distribution and 3 Applying polydispersity. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-[illegible] molecular weight distribution and 3 Applying polydispersity. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Derive the mathematical expressions Mn& Mw 3 Understanding Summarize the various experimental methods

Derive the mathematical expressions Mn& Mw 3 Understanding Summarize the various experimental methods is an official topic in Module 2 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Derive the mathematical expressions Mn& Mw 3 Understanding Summarize the various experimental methods using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, derive the mathematical expressions mn& mw 3 understanding summarize the various experimental methods should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Derive the mathematical expressions Mn& Mw 3 Understanding Summarize the various experimental methods means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Derive the mathematical expressions Mn& Mw 3 Understanding Summarize the various experimental methods [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

for the determination of molecular weight of 4 Understanding polymers Molecular weight of polymers - Average molecular weight concept - different molecular weight of polymers - Distinguish molecular weight of micro and macro polymers - Different molecular weights of polymers - Mn, Mw, Mv and Mz-Derive equations for Mn and Mw. Polydispersity and molecular weight distribution. Determination of Mn and Mw - End group analysis - Ebulliometer, Cryoscopy, Light scattering and Gel permeation chromatography, Viscometry methods.

for the determination of molecular weight of 4 Understanding polymers Molecular weight of polymers - Average molecular weight concept - different molecular weight of polymers - Distinguish molecular weight of micro and macro polymers - Different molecular weights of polymers - Mn, Mw, Mv and Mz-Derive equations for Mn and Mw. Polydispersity and molecular weight distribution. Determination of Mn and Mw - End group analysis - Ebulliometer, Cryoscopy, Light scattering and Gel permeation chromatography, Viscometry methods. is an official topic in Module 2 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify for the determination of molecular weight of 4 Understanding polymers Molecular weight of polymers - Average molecular weight concept - different molecular weight of polymers - Distinguish molecular weight of micro and macro polymers - Different molecular weights of polymers - Mn, Mw, Mv and Mz-Derive equations for Mn and Mw. Polydispersity and molecular weight distribution. Determination of Mn and Mw - End group analysis - Ebulliometer, Cryoscopy, Light scattering and Gel permeation chromatography, Viscometry methods. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, for the determination of molecular weight of 4 understanding polymers molecular weight of polymers - average molecular weight concept - different molecular weight of polymers - distinguish molecular weight of micro and macro polymers - different molecular weights of polymers - mn, mw, mv and mz-derive equations for mn and mw. polydispersity and molecular weight distribution. determination of mn and mw - end group analysis - ebulliometer, cryoscopy, light scattering and gel permeation chromatography, viscometry methods. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What for the determination of molecular weight of 4 Understanding polymers Molecular weight of polymers - Average molecular weight concept - different molecular weight of polymers - Distinguish molecular weight of micro and macro polymers - Different molecular weights of polymers - Mn, Mw, Mv and Mz-Derive equations for Mn and Mw. Polydispersity and molecular weight distribution. Determination of Mn and Mw - End group analysis - Ebulliometer, Cryoscopy, Light scattering and Gel permeation chromatography, Viscometry methods. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ for the determination of molecular weight of 4 Understanding polymers Molecular weight of polymers - Average molecular weight concept - different molecular weight of polymers - Distinguish molecular weight of micro and macro polymers - Different molecular weights of polymers - Mn, Mw, Mv and Mz-Derive equations for Mn and Mw. Polydispersity and molecular weight distribution. Determination of Mn and Mw - End group analysis - Ebulliometer, Cryoscopy, Light scattering and Gel permeation chromatography, Viscometry methods. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
determination
            Interpret the Concept of molecular weight of
  M2.01                                                           2
Applying
            micro and macro molecules.
            Explain different molecular weight of polymers,
  M2.02     molecular weight distribution and                     3
Applying
            polydispersity.
  M2.03     Derive the mathematical expressions Mn& Mw            3
Understanding
             Summarize the various experimental methods
  M2.04      for the determination of molecular weight of           4
Understanding
             polymers
             Series Test - I                                        1

 Contents:
 Molecular weight of polymers - Average molecular weight concept - different
molecular
 weight of polymers - Distinguish molecular weight of micro and macro polymers -
 Different molecular weights of polymers - Mn, Mw, Mv and Mz-Derive equations for
Mn
 and Mw. Polydispersity and molecular weight distribution. Determination of Mn
and Mw -
 End group analysis - Ebulliometer, Cryoscopy, Light scattering and Gel
permeation
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Illustrate the stereo regularity of polymers

This module is presented from the official syllabus scope for Polymer Science. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Illustrate the stereo regularity of polymers
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

2 Understanding and geometrical structures Explain Optical and geometrical isomerism in

2 Understanding and geometrical structures Explain Optical and geometrical isomerism in is an official topic in Module 3 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding and geometrical structures Explain Optical and geometrical isomerism in using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding and geometrical structures explain optical and geometrical isomerism in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding and geometrical structures Explain Optical and geometrical isomerism in means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 2 Understanding and geometrical structures Explain Optical and geometrical isomerism in □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Understanding polymers Comprehend the Importance of Ziegler Natta

2 Understanding polymers Comprehend the Importance of Ziegler Natta is an official topic in Module 3 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding polymers Comprehend the Importance of Ziegler Natta using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding polymers comprehend the importance of ziegler natta should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding polymers Comprehend the Importance of Ziegler Natta means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-□□ 2 Understanding polymers Comprehend the Importance of Ziegler Natta □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

3 Understanding catalyst in stereoregularity of polymers

3 Understanding catalyst in stereoregularity of polymers is an official topic in Module 3 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding catalyst in stereoregularity of polymers using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding catalyst in stereoregularity of polymers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding catalyst in stereoregularity of polymers means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] 3 Understanding catalyst in stereoregularity of polymers [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain the microstructures of PP,PB,CR,IR 4 Understanding Stereo regularity of polymers - Polymer microstructure - chemical and Geometrical structure - Stereo regular polymer - Optical and geometrical isomerism. Ziegler Natta catalyst - typical combinations. Tacticity - isotactic, syndiotactic, atactic. Microstructure - BR, IR, CR - Cis, Trans, 1, 2 additions, 3, 4 addition and tactic structures.

Explain the microstructures of PP,PB,CR,IR 4 Understanding Stereo regularity of polymers - Polymer microstructure - chemical and Geometrical structure - Stereo regular polymer - Optical and geometrical isomerism. Ziegler Natta catalyst - typical combinations. Tacticity - isotactic, syndiotactic, atactic. Microstructure - BR, IR, CR - Cis, Trans, 1, 2 additions, 3, 4 addition and tactic structures. is an official topic in Module 3 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the microstructures of PP,PB,CR,IR 4 Understanding Stereo regularity of polymers - Polymer microstructure - chemical and Geometrical structure - Stereo regular polymer - Optical and geometrical isomerism. Ziegler Natta catalyst - typical combinations. Tacticity - isotactic, syndiotactic, atactic. Microstructure - BR, IR, CR - Cis, Trans, 1, 2 additions, 3, 4 addition and tactic structures. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the microstructures of pp,pb,cr,ir 4 understanding stereo regularity of polymers - polymer microstructure - chemical and geometrical structure - stereo regular polymer - optical and geometrical isomerism. ziegler natta catalyst - typical combinations. tacticity - isotactic, syndiotactic, atactic. microstructure - br, ir, cr - cis, trans, 1, 2 additions, 3, 4 addition and tactic structures. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the microstructures of PP,PB,CR,IR 4 Understanding Stereo regularity of polymers - Polymer microstructure - chemical and Geometrical structure - Stereo regular polymer - Optical and geometrical isomerism. Ziegler Natta catalyst - typical combinations. Tacticity - isotactic, syndiotactic, atactic. Microstructure - BR, IR, CR - Cis, Trans, 1, 2 additions, 3, 4 addition and tactic structures. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Explain the microstructures of PP,PB,CR,IR 4 Understanding Stereo regularity of polymers - Polymer microstructure - chemical and Geometrical structure - Stereo regular polymer - Optical and geometrical isomerism. Ziegler Natta catalyst - typical combinations. Tacticity - isotactic, syndiotactic, atactic. Microstructure - BR, IR, CR - Cis, Trans, 1, 2 additions, 3, 4 addition and tactic structures. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Illustrate the stereo regularity of polymers
             Mention the microstructures based on chemical
  M3.01                                                         2
Understanding
             and geometrical structures
             Explain Optical and geometrical isomerism in
  M3.02                                                         2
Understanding
             polymers
             Comprehend the Importance of Ziegler Natta
  M3.03                                                         3
Understanding
             catalyst in stereoregularity of polymers

  M3.04      Explain the microstructures of PP,PB,CR,IR         4
Understanding

Contents:
Stereo regularity of polymers - Polymer microstructure - chemical and Geometrical
structure
- Stereo regular polymer - Optical and geometrical isomerism. Ziegler Natta
catalyst - typical
combinations. Tacticity - isotactic, syndiotactic, atactic. Microstructure - BR,
IR, CR - Cis,
Trans, 1, 2 additions, 3, 4 addition and tactic structures.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Explain Polymer reactions and polymer modifications

This module is presented from the official syllabus scope for Polymer Science. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain Polymer reactions and polymer modifications
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

1 Understanding polymers and simple molecules Illustrate the effect of hydroxyl ,carboxylic and

1 Understanding polymers and simple molecules Illustrate the effect of hydroxyl ,carboxylic and is an official topic in Module 4 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding polymers and simple molecules Illustrate the effect of hydroxyl ,carboxylic and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding polymers and simple molecules illustrate the effect of hydroxyl ,carboxylic and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding polymers and simple molecules Illustrate the effect of hydroxyl ,carboxylic and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 1 Understanding polymers and simple molecules Illustrate the effect of hydroxyl ,carboxylic and □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

1 Understanding aldehydic groups in polymer Reactions Explain acidolysis, hydrolysis, aminolysis

1 Understanding aldehydic groups in polymer Reactions Explain acidolysis, hydrolysis, aminolysis is an official topic in Module 4 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding aldehydic groups in polymer Reactions Explain acidolysis, hydrolysis, aminolysis using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding aldehydic groups in polymer reactions explain acidolysis, hydrolysis, aminolysis should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding aldehydic groups in polymer Reactions Explain acidolysis, hydrolysis, aminolysis means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 1 Understanding aldehydic groups in polymer Reactions Explain acidolysis, hydrolysis, aminolysis □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

addition and substitution reactions ,and 4 Understanding crosslinking reactions in polymers Summarize the physical and chemical methods

addition and substitution reactions ,and 4 Understanding crosslinking reactions in polymers Summarize the physical and chemical methods is an official topic in Module 4 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify addition and substitution reactions ,and 4 Understanding crosslinking reactions in polymers Summarize the physical and chemical methods using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, addition and substitution reactions ,and 4 understanding crosslinking reactions in polymers summarize the physical and chemical methods should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What addition and substitution reactions ,and 4 Understanding crosslinking reactions in polymers Summarize the physical and chemical methods means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ addition and substitution reactions ,and 4 Understanding crosslinking reactions in polymers Summarize the physical and chemical methods □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

4 Understanding of polymer modifications with applications Differences between reactions of polymers and simple molecules - effect of functional groups in polymers - Hydroxyl - carboxylic - aldehydic groups. Hydrolysis - Acidolysis - Aminolysis -Hydrogenation - Addition, Substitution - cross linking reactions, sulphur, peroxide, metal oxide. Polymer modification - Physical methods. Blends - Alloys Composites. Chemical methods- co polymerization - Grafting – IPN. Text / Reference: T/R Book Title/Author Polymer science V.R.Gowarikar (Publisher - New Age International (P) Ltd, New R1 Delhi- 2005 Edition 3. Text Book of Polymer Science - Billmeyer. J.R (John Wiley & Sons (Aia) PTE Ltd, R2 Singapore, 2008 Edition Introductory Polymer Chemistry - G.S.Misra (Publisher - New Age International R3 (P) Ltd, New Delhi - 2005 Edition Principles of Polymer Science - P.Bahadur, N.V.Sastri (Publisher - Alpha Science R4 International Ltd, 2005 Edition Online Resources: Sl.No Website Link 1 https://books.google.com/books/about/Polymer_Science.html?id=mvCzE 2 libguides.uml.edu/polymer science

4 Understanding of polymer modifications with applications Differences between reactions of polymers and simple molecules - effect of functional groups in polymers - Hydroxyl - carboxylic - aldehydic groups. Hydrolysis - Acidolysis - Aminolysis -Hydrogenation - Addition, Substitution - cross linking reactions, sulphur, peroxide, metal oxide. Polymer modification - Physical methods. Blends - Alloys Composites. Chemical methods- co polymerization - Grafting – IPN. Text / Reference: T/R Book Title/Author Polymer science V.R.Gowarikar (Publisher - New Age International (P) Ltd, New R1 Delhi- 2005 Edition 3. Text Book of Polymer Science - Billmeyer. J.R (John Wiley & Sons (Aia) PTE Ltd, R2 Singapore, 2008 Edition Introductory Polymer Chemistry - G.S.Misra (Publisher - New Age International R3 (P) Ltd, New Delhi - 2005 Edition Principles of Polymer Science - P.Bahadur, N.V.Sastri (Publisher - Alpha Science R4 International Ltd, 2005 Edition Online Resources: Sl.No Website Link 1 https://books.google.com/books/about/Polymer_Science.html?id=mvCzE 2 libguides.uml.edu/polymer science is an official topic in Module 4 of Polymer Science. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding of polymer modifications with applications Differences between reactions of polymers and simple molecules - effect of functional groups in polymers - Hydroxyl - carboxylic - aldehydic groups. Hydrolysis - Acidolysis - Aminolysis -Hydrogenation - Addition, Substitution - cross linking reactions, sulphur, peroxide, metal oxide. Polymer modification - Physical methods. Blends - Alloys Composites. Chemical methods- co polymerization - Grafting – IPN. Text / Reference: T/R Book Title/Author Polymer science V.R.Gowarikar (Publisher - New Age International (P) Ltd, New R1 Delhi- 2005 Edition 3. Text Book of Polymer Science - Billmeyer. J.R (John Wiley & Sons (Aia) PTE Ltd, R2 Singapore, 2008 Edition Introductory Polymer Chemistry - G.S.Misra (Publisher - New Age International R3 (P) Ltd, New Delhi - 2005 Edition Principles of Polymer Science - P.Bahadur, N.V.Sastri (Publisher - Alpha Science R4 International Ltd, 2005 Edition Online Resources: Sl.No Website Link 1 https://books.google.com/books/about/Polymer_Science.html?id=mvCzE 2 libguides.uml.edu/polymer science using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding of polymer modifications with applications differences between reactions of polymers and simple molecules - effect of functional groups in polymers - hydroxyl - carboxylic - aldehydic groups. hydrolysis - acidolysis - aminolysis -hydrogenation - addition, substitution - cross linking reactions, sulphur, peroxide, metal oxide. polymer modification - physical methods. blends - alloys composites. chemical methods- co polymerization - grafting – ipn. text / reference: t/r book title/author polymer science v.r.gowarikar (publisher - new age international (p) ltd, new r1 delhi- 2005 edition 3. text book of polymer science - billmeyer. j.r (john wiley & sons (aia) pte ltd, r2 singapore, 2008 edition introductory polymer chemistry - g.s.misra (publisher - new age international r3 (p) ltd, new delhi - 2005 edition principles of polymer science - p.bahadur, n.v.sastri (publisher - alpha science r4 international ltd, 2005 edition online resources: sl.no website link 1 https://books.google.com/books/about/polymer_science.html?id=mvcze 2 libguides.uml.edu/polymer science should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding of polymer modifications with applications Differences between reactions of polymers and simple molecules - effect of functional groups in polymers - Hydroxyl - carboxylic - aldehydic groups. Hydrolysis - Acidolysis - Aminolysis -Hydrogenation - Addition, Substitution -

Aspect	What to prepare
	cross linking reactions, sulphur, peroxide, metal oxide. Polymer modification - Physical methods. Blends - Alloys Composites. Chemical methods- co polymerization - Grafting – IPN. Text / Reference: T/R Book Title/Author Polymer science V.R.Gowarikar (Publisher - New Age International (P) Ltd, New R1 Delhi- 2005 Edition 3. Text Book of Polymer Science - Billmeyer. J.R (John Wiley & Sons (Aia) PTE Ltd, R2 Singapore, 2008 Edition Introductory Polymer Chemistry - G.S.Misra (Publisher - New Age International R3 (P) Ltd, New Delhi - 2005 Edition Principles of Polymer Science - P.Bahadur, N.V.Sastri (Publisher - Alpha Science R4 International Ltd, 2005 Edition Online Resources: Sl.No Website Link 1 https://books.google.com/books/about/Polymer_Science.html?id=mvCzE 2 libguides.uml.edu/polymer science means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 4 Understanding of polymer modifications with applications Differences between reactions of polymers and simple molecules - effect of functional groups in polymers - Hydroxyl - carboxylic - aldehydic groups. Hydrolysis - Acidolysis - Aminolysis -Hydrogenation - Addition, Substitution - cross linking reactions, sulphur, peroxide, metal oxide. Polymer modification - Physical methods. Blends - Alloys Composites. Chemical methods- co polymerization - Grafting – IPN. Text / Reference: T/R Book Title/Author Polymer science V.R.Gowarikar (Publisher - New Age International (P) Ltd, New R1 Delhi- 2005 Edition 3. Text Book of Polymer Science - Billmeyer. J.R (John Wiley & Sons (Aia) PTE Ltd, R2 Singapore, 2008 Edition Introductory Polymer Chemistry - G.S.Misra (Publisher - New Age International R3 (P) Ltd, New Delhi - 2005 Edition Principles of Polymer Science - P.Bahadur, N.V.Sastri (Publisher - Alpha Science R4 International Ltd, 2005 Edition Online Resources: Sl.No Website Link 1 https://books.google.com/books/about/Polymer_Science.html?id=mvCzE 2 libguides.uml.edu/polymer science □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Explain Polymer reactions and polymer modifications

             Identify the differences between reactions of
  M4.01                                                       1
Understanding
             polymers and simple molecules
             Illustrate the effect of hydroxyl ,carboxylic and
  M4.02                                                       1
Understanding
             aldehydic groups in polymer Reactions
             Explain acidolysis, hydrolysis, aminolysis
  M4.03      addition and substitution reactions ,and         4
Understanding
             crosslinking reactions in polymers
             Summarize the physical and chemical methods
  M4.05                                                       4
Understanding
             of polymer modifications with applications
             Series Test - II                                 1

Contents:
Differences between reactions of polymers and simple molecules - effect of
functional
groups in polymers - Hydroxyl - carboxylic - aldehydic groups. Hydrolysis -
Acidolysis -
Aminolysis -Hydrogenation - Addition, Substitution - cross linking reactions,
sulphur,
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain basis of States of phase and aggregates of 3 Understanding polymers. Explain Glass transition temperature (Tg)-. (3 marks)

Answer: Begin with the standard definition or identity of *basis of States of phase and aggregates of 3 Understanding polymers. Explain Glass transition temperature (Tg)-*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Melting temperature (Tm) -Flow temperature 2 Understanding (Tf) Explain Crystalline polymer and amorphous. (3 marks)

Answer: Begin with the standard definition or identity of *Melting temperature (Tm) - Flow temperature 2 Understanding (Tf) Explain Crystalline polymer and amorphous*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding polymer Explain Polymer identification by chemical and. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding polymer Explain Polymer identification by chemical and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Applying physical methods State of phase and aggregates of polymers and polymer identification Glass transition temperature (Tg) - Melting temperature (Tm) - Flow temperature (Tf). Crystalline polymer - Amorphous polymer - Crystallinity Crystallisability -factors affecting Tg - Dilatometry determination of Tg-X-Ray diffraction method for determination of crystallinity - Comparison of Elastomers, plastics and fibers in terms of Tg and crystallinity. Polymer identification - chemical and physical methods. TGA, DSC and IR spectroscopy. Explain the concept of molecular weight and experimental method of its. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying physical methods State of phase and aggregates of polymers and polymer identification Glass transition temperature (Tg) - Melting temperature (Tm) - Flow temperature (Tf). Crystalline polymer - Amorphous polymer - Crystallinity Crystallisability -factors affecting Tg - Dilatometry determination of Tg-X-Ray diffraction method for determination of crystallinity - Comparison of Elastomers, plastics and fibers in terms of Tg and crystallinity. Polymer identification - chemical and physical methods. TGA, DSC and IR spectroscopy. Explain the concept of molecular weight and experimental method of its*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Module 2

Define or explain 2 Applying micro and macro molecules. Explain different molecular weight of polymers,. (3 marks)

Answer: Begin with the standard definition or identity of *2 Applying micro and macro molecules. Explain different molecular weight of polymers,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain molecular weight distribution and 3 Applying polydispersity.. (3 marks)

Answer: Begin with the standard definition or identity of *molecular weight distribution and 3 Applying polydispersity.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps,

□□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Derive the mathematical expressions Mn& Mw 3 Understanding Summarize the various experimental methods. (3 marks)

Answer: Begin with the standard definition or identity of *Derive the mathematical expressions Mn& Mw 3 Understanding Summarize the various experimental methods*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain for the determination of molecular weight of 4 Understanding polymers Molecular weight of polymers - Average molecular weight concept - different molecular weight of polymers - Distinguish molecular weight of micro and macro polymers - Different molecular weights of polymers - Mn, Mw, Mv and Mz-Derive equations for Mn and Mw. Polydispersity and molecular weight distribution. Determination of Mn and Mw - End group analysis - Ebulliometer, Cryoscopy, Light scattering and Gel permeation chromatography, Viscometry methods.. (3 marks)

Answer: Begin with the standard definition or identity of *for the determination of molecular weight of 4 Understanding polymers Molecular weight of polymers - Average molecular weight concept - different molecular weight of polymers - Distinguish molecular weight of micro and macro polymers - Different molecular weights of polymers - Mn, Mw, Mv and Mz-Derive equations for Mn and Mw. Polydispersity and molecular weight distribution. Determination of Mn and Mw - End group analysis - Ebulliometer, Cryoscopy, Light scattering and Gel permeation chromatography, Viscometry methods.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Module 3

Define or explain 2 Understanding and geometrical structures Explain Optical and geometrical isomerism in. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding and geometrical structures Explain Optical and geometrical isomerism in*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding polymers Comprehend the Importance of Ziegler Natta. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding polymers Comprehend the Importance of Ziegler Natta*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding catalyst in stereoregularity of polymers. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding catalyst in stereoregularity of polymers*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain the microstructures of PP,PB,CR,IR 4 Understanding Stereo regularity of polymers - Polymer microstructure - chemical and Geometrical structure - Stereo regular polymer - Optical and geometrical

isomerism. Ziegler Natta catalyst - typical combinations. Tacticity - isotactic, syndiotactic, atactic. Microstructure - BR, IR, CR - Cis, Trans, 1, 2 additions, 3, 4 addition and tactic structures.. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the microstructures of PP,PB,CR,IR 4 Understanding Stereo regularity of polymers - Polymer microstructure - chemical and Geometrical structure - Stereo regular polymer - Optical and geometrical isomerism. Ziegler Natta catalyst - typical combinations. Tacticity - isotactic, syndiotactic, atactic. Microstructure - BR, IR, CR - Cis, Trans, 1, 2 additions, 3, 4 addition and tactic structures.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain 1 Understanding polymers and simple molecules Illustrate the effect of hydroxyl ,carboxylic and. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding polymers and simple molecules Illustrate the effect of hydroxyl ,carboxylic and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 1 Understanding aldehydic groups in polymer Reactions Explain acidolysis, hydrolysis, aminolysis. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding aldehydic groups in polymer Reactions Explain acidolysis, hydrolysis, aminolysis*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain addition and substitution reactions ,and 4 Understanding crosslinking reactions in polymers Summarize the physical and chemical methods. (3 marks)

Answer: Begin with the standard definition or identity of *addition and substitution reactions ,and 4 Understanding crosslinking reactions in polymers Summarize the physical and chemical methods*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding of polymer modifications with applications Differences between reactions of polymers and simple molecules - effect of functional groups in polymers - Hydroxyl - carboxylic - aldehydic groups. Hydrolysis - Acidolysis - Aminolysis -Hydrogenation - Addition, Substitution - cross linking reactions, sulphur, peroxide, metal oxide. Polymer modification - Physical methods. Blends - Alloys Composites. Chemical methods- co polymerization - Grafting – IPN. Text / Reference: T/R Book Title/Author Polymer science V.R.Gowarikar (Publisher - New Age International (P) Ltd, New R1 Delhi- 2005 Edition 3. Text Book of Polymer Science - Billmeyer. J.R (John Wiley & Sons (Aia) PTE Ltd, R2 Singapore, 2008 Edition Introductory Polymer Chemistry - G.S.Misra (Publisher - New Age International R3 (P) Ltd, New Delhi - 2005 Edition Principles of Polymer Science - P.Bahadur, N.V.Sastri (Publisher - Alpha Science R4 International Ltd, 2005 Edition Online Resources: Sl.No Website Link 1 https://books.google.com/books/about/Polymer_Science.html?id=mvCzE 2 libguides.uml.edu/polymer science. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding of polymer modifications with applications Differences between reactions of polymers and simple molecules - effect of functional groups in polymers - Hydroxyl - carboxylic - aldehydic groups. Hydrolysis - Acidolysis - Aminolysis -Hydrogenation - Addition, Substitution - cross linking reactions, sulphur, peroxide, metal oxide. Polymer modification - Physical methods. Blends - Alloys Composites. Chemical methods- co polymerization - Grafting –*

IPN. Text / Reference: T/R Book Title/Author Polymer science V.R.Gowarikar (Publisher - New Age International (P) Ltd, New R1 Delhi- 2005 Edition 3. Text Book of Polymer Science - Billmeyer. J.R (John Wiley & Sons (Aia) PTE Ltd, R2 Singapore, 2008 Edition Introductory Polymer Chemistry - G.S.Misra (Publisher - New Age International R3 (P) Ltd, New Delhi - 2005 Edition Principles of Polymer Science - P.Bahadur, N.V.Sastri (Publisher - Alpha Science R4 International Ltd, 2005 Edition Online Resources: Sl.No Website Link 1 https://books.google.com/books/about/Polymer_Science.html?id=mvCzE 2 libguides.uml.edu/polymer science. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **basis of States of phase and aggregates of 3 Understanding polymers. Explain Glass transition temperature (Tg)-**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **2 Applying micro and macro molecules. Explain different molecular weight of polymers,**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **2 Understanding and geometrical structures Explain Optical and geometrical isomerism in**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **1 Understanding polymers and simple molecules Illustrate the effect of hydroxyl ,carboxylic and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.

- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- https://books.google.com/books/about/Polymer_Science.html?id=mvCzE libguides.uml.edu/polymer science

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 3092. Verify institutional updates against the current official curriculum.